Ishikawa Iteration and Common Fixed Point Theorems

Chapter 5: Fixed Point Theorems (Pages 405-428)

Pathak's Fixed Point Theory

2024

Course Overview

What are Fixed Point Iteration Methods? I

Definition: A fixed point iteration method is a sequence of approximations to solve equations of the form:

$$x = T(x)$$

where $T : X \to X$ is a mapping on some space X.

Key Question: How do we find solutions to operator equations?

Answer: Using iteration schemes:

- **One-step (Picard):** $x_{n+1} = Tx_n$
- **Two-step (Ishikawa/Mann):** Involve multiple stages per iteration
- **Multi-step:** Generalize to families of mappings

Why use two-step methods?

- More flexible convergence conditions
- Applicable to broader classes of mappings

What are Fixed Point Iteration Methods? II

- Better stability properties
- Can handle nonexpansive mappings

Common Fixed Points: Definitions

Definition 5.66 (George et al.): Let (X, d) be a metric space, k a positive integer, $T : X^k \to X$ and $f : X \to X$ be mappings.

(a) Coincidence Point:

$$x \in X \text{ is a coincidence point of } f \text{ and } T \iff f(x) = T(x, x, \ldots, x)$$

(b) Common Fixed Point:

$$x \text{ is a common fixed point } \iff x = f(x) = T(x, x, \ldots, x)$$

(c) Commuting Mappings:

$$f(T(x, x, \ldots, x)) = T(f(x), f(x), \ldots, f(x)) \quad \forall x \in X$$

(d) Weakly Compatible: Mappings commute at their coincidence points.

Theorem 5.118: Let (X, d) be a b-metric space with $s \geq 1$. For any positive integer k, let $f : X^k \to X$ and $g : X \to X$ satisfy:

Conditions:

(i) $f(X^k) \subseteq g(X)$

(ii) $d(f(x_1, \ldots, x_k), f(x_2, \ldots, x_{k+1})) \leq \lambda\phi(d(gx_1, gx_2), \ldots, d(gx_k, gx_{k+1}))$ where $\lambda \in (0, 1/s^k)$

(iii) $d(f(u, \ldots, u), f(v, \ldots, v)) < d(gu, gv)$ for all $u, v \in X$

(iv) $g(X)$ is complete

Conclusion:

- f and g have a unique common fixed point
- The sequence $\{y_n\}$ defined by $y_n = g(x_n) = f(x_n, x_{n+1}, \ldots, x_{n+k-1})$ converges to the common fixed point

Definition 5.67: A mapping $F : K \to K$ on a convex subset of a normed linear space is:

$$\textbf{Affine} \iff F(\alpha x + (1-\alpha)y) = \alpha F(x) + (1-\alpha)F(y), \quad \alpha \in (0,1)$$

Theorem 5.120 (Markov-Kakutani): Let K be a compact convex subset of a normed linear space X. If $\mathcal{F}$ is a commuting family of continuous affine mappings of K into itself, then there exists $u \in K$ such that $Fu = u$ for all $F \in \mathcal{F}$.

Applications:

- Solving systems of equations
- Equilibrium problems
- Game theory

Theorem 5.123: de Marr's Theorem (Nonexpansive Mappings)

Theorem 5.123 (de Marr, 1963): Let K be a compact convex subset of a Banach space X, and let $\mathcal{F}$ be a family of commuting nonexpansive mappings of K into itself. Then there is a common fixed point for the family.

Definition: A mapping $T : K \to K$ is **nonexpansive** if:

$$\|Tx - Ty\| \leq \|x - y\| \quad \forall x, y \in K$$

Key Properties of Nonexpansive Mappings:

- More general than contractions
- No Lipschitz constant less than 1 required
- May not have unique fixed points
- Still have nice convergence properties under iteration

Related Results:

- **Theorem 5.126 (Browder):** Nonexpansive families have common fixed points in uniformly convex Banach spaces
- **Theorem 5.124-5.125:** Variations for weakly compact and bounded convex sets

Iteration Schemes: General Framework

Problem: Find a common fixed point of a family $\{F_i\}_{i=1}^k$ of nonexpansive mappings.

Approach: Define an iteration scheme using weighted averages:

For $0 < \alpha < 1$, define operators:

$$
\begin{aligned}
U_0 &= I \quad \text{(identity mapping)} \\
U_r &= (1-\alpha)I + \alpha F_r U_{r-1}, \quad r = 1, 2, \ldots, k
\end{aligned}
$$

The iteration scheme generates a sequence by repeatedly applying U_k:

$$\{U_k^n(x)\}_{n=1}^{\infty}$$

Interpretation:

- Each U_r is a convex combination of identity and the mapped previous operator
- The combined operator U_k applies all k mappings in sequence
- Linearly weighted by parameter α

Theorem 5.130: Strong Convergence

Theorem 5.130: Let K be a compact convex subset of a strictly convex Banach space X, and $\{F_i\}_{i=1}^k$ a finite family of nonexpansive self-mappings of K with nonempty common fixed point set.

Then for any $x \in K$, the sequence $\{U_k^n(x)\}_{n=1}^\infty$ **converges strongly** to a common fixed point of $\{F_i\}_{i=1}^k$.

Proof Strategy:

1. Show U_k is nonexpansive: $\|U_k(x) - U_k(y)\| \leq \|x - y\|$
2. Use Edelstein's theorem for contraction properties
3. Exploit strict convexity for strong convergence
4. Bounded sequences in finite dimensions converge

Key Requirement: Strict convexity of the space (e.g., Hilbert spaces, L^p spaces with $p > 1$)

Theorem 5.131: Weak Convergence

Theorem 5.131: If X is a uniformly convex Banach space satisfying Opial's condition (e.g., Hilbert spaces), let K be closed and convex, and $\{F_i\}_{i=1}^k$ a family of nonexpansive self-mappings of K with nonempty common fixed point set.

Then for any $x \in K$, the sequence $\{U_k^n(x)\}_{n=1}^\infty$ **converges weakly** to a common fixed point.

Opial's Condition: For any sequence $\{x_n\}$ and $x \in X$:

$$\limsup_{n\to\infty} \|x_n - x\| < \limsup_{n\to\infty} \|x_n - y\|, \quad \text{whenever } x_n \rightharpoonup x \text{ and } x \neq y$$

Examples of spaces satisfying Opial's condition:

- All Hilbert spaces
- ℓ^p spaces for $1 \leq p < \infty$
- But NOT L^p spaces for $p \neq 2$

Theorem 5.132: Parameter-Controlled Convergence

Theorem 5.132: Let K be closed and convex in a uniformly convex Banach space X with Fréchet differentiable norm. Let $\{F_i\}_{i=1}^k$ be nonexpansive self-mappings with nonempty fixed point set, and $\{c_n\}$ a sequence with:

Conditions on sequence $\{c_n\}$:

(i) $0 \leq c_n \leq 1$

(ii) $\sum_{n=1}^{\infty} c_n(1 - c_n) = \infty$

Define the iteration:

$$x_{n+1} = (1 - c_n)x_n + c_n F_k U_{k-1} x_n, \quad n \geq 1$$

Then $\{x_n\}$ **converges weakly** to a common fixed point of $\{F_i\}_{i=1}^k$.

Interpretation:

- Parameter c_n controls step size at each iteration
- Condition (ii) ensures sufficient exploration (not too greedy)
- Relates to gradient descent and optimization methods

Example parameters: $c_n = 1/(n+1)$, $c_n = 1/\sqrt{n}$

Application: System of Operator Equations I

Problem: Solve a system of operator equations:

$$x - T_i x = f_i, \quad i = 1, 2, \ldots, k$$

where T_i are nonexpansive and f_i are given elements.

Solution Approach: Define the family of mappings:

$$F_i(x) = f_i + T_i(x), \quad i = 1, 2, \ldots, k$$

Key Observation:

- x is a solution to the system $\iff$ x is a common fixed point of $\{F_i\}_{i=1}^{k}$
- Each F_i is nonexpansive (since T_i is nonexpansive)
- Apply Theorems 5.130, 5.131, or 5.132 to find the solution iteratively

Application: System of Operator Equations II

Algorithm (Theorem 5.132):

```
x = x_initial
for n = 1, 2, ...:
  # Compute weighted averages
  U = I
  for r = 1, ..., k:
    U = (1 − α)I + αF_r U
  # Apply parameter-controlled step
  c_n = 1/(n + 1)
  x = (1 − c_n)x + c_n F_k U x
  if converged(x): break
```

Fixed Points of Contraction Sequences

Question: If $\{F_n\}$ is a sequence of contractions converging to F, do their fixed points converge to a fixed point of F?

Answer: YES, under appropriate conditions!

Theorem 5.133 (Bonsall, 1962): Let (X, d) be a complete metric space, and let T and T_n $(n = 1, 2, \ldots)$ be contraction mappings with the same Lipschitz constant $k < 1$ and fixed points u and u_n respectively.

If $\lim_{n\to\infty} T_n x = Tx$ for every $x \in X$, then $\lim_{n\to\infty} u_n = u$.

Proof Sketch:

1. Use Banach contraction principle: $d(u_n, T_n^m x_0) \leq \frac{k^m}{1-k} d(T_n x_0, x_0)$
2. Set $m = 0$ and $x_0 = u$: $d(u_n, u) \leq \frac{1}{1-k} d(T_n u, u)$
3. Take limit as $n \to \infty$: $\lim_{n\to\infty} d(u_n, u) = 0$

Applications:

- Perturbation analysis
- Stability of numerical schemes
- Continuity of solutions under parameter changes

Multivalued Mappings and Hausdorff Metric

Definition: For nonempty closed subsets of X, the Hausdorff metric is:

$$H(A, B) = \max\{\sup_{x \in A} d(x, B), \sup_{y \in B} d(y, A)\}$$

Theorem 5.127 (Bose-Mukerjee): Let (X, d) be a complete bounded metric space, and $F_i : X \to \mathrm{CL}(X)$ (multivalued mappings to closed sets) satisfy:

$$H(F_1(x), F_2(y)) \leq a_1 d(x, F_1(x)) + a_2 d(y, F_2(y)) + a_3 d(y, F_1(x)) + a_4 d(x, F_2(y)) + a_5 d(x, y)$$

where $\sum_{i=1}^{5} a_i < 1$ and $a_1 = a_2$ or $a_3 = a_4$.

Then F_1 and F_2 have a common fixed point.

Key Features:

- Generalizes Kannan-Reich condition to multivalued mappings
- Allows non-contractive mappings
- Oscillating (alternating) indices i, j in sequence
- Distance to set (not just point) appears in condition

Ishikawa Iteration: Definition

Definition: The Ishikawa iteration for a mapping $T : K \to K$ on a convex subset K of a Banach space is defined as follows:

For a given $x_0 \in K$ and sequences $\{\alpha_n\}$, $\{\beta_n\}$ in $(0, 1)$:

Two-step scheme:

$$y_n = (1 - \alpha_n)x_n + \alpha_n Tx_n \tag{1}$$

$$x_{n+1} = (1 - \beta_n)x_n + \beta_n Ty_n \tag{2}$$

Notation:

- α_n, β_n: step-size parameters (usually in $(0, 1)$)
- y_n: intermediate iterate (first step)
- x_{n+1}: next iterate (second step)
- T: the nonexpansive mapping

Special Cases:

- $\alpha_n = 0$: Reduces to Mann iteration (with parameter β_n)
- $\alpha_n = \beta_n = 1$: Reduces to Picard iteration $x_{n+1} = T(x_n)$

Convergence of Ishikawa Iteration

Theorem (Ishikawa, 1974): Let K be a closed, bounded, convex subset of a uniformly convex Banach space X. Let $T : K \to K$ be a nonexpansive mapping. Then for sequences $\{\alpha_n\}$ and $\{\beta_n\}$ with:

Conditions:

1. $0 < a \leq \alpha_n, \beta_n \leq b < 1$ for all n
2. $\sum_{n=1}^{\infty} \alpha_n = \infty$ and $\sum_{n=1}^{\infty} \beta_n = \infty$

The Ishikawa sequence $\{x_n\}$ **converges weakly** to a fixed point of T.

Key Advantages:

- Works for nonexpansive mappings (not just contractions)
- More flexibility in choosing parameters
- Applicable to fixed point problems in infinite-dimensional spaces
- Better numerical stability in some cases

Comparison with Picard:

- Picard: $x_{n+1} = Tx_n$ (simpler, faster for contractions)
- Ishikawa: Two-step (more robust for nonexpansive maps)

Mann Iteration (Special Case)

Definition: The Mann iteration is obtained from Ishikawa with $\alpha_n = 0$:

$$x_{n+1} = (1 - \beta_n)x_n + \beta_n Tx_n = x_n + \beta_n(Tx_n - x_n)$$

Interpretation:

- No intermediate step, direct convex combination
- β_n controls how much we move toward Tx_n
- β_n small $\Rightarrow$ slow convergence but stable
- β_n large $\Rightarrow$ faster but potentially unstable

Convergence (Mann): Under similar conditions, Mann iteration also converges weakly to a fixed point.

Practical Choices:

- Constant: $\beta_n = \beta \in (0, 1)$ for all n (simple, often $\beta = 0.5$)
- Decreasing: $\beta_n = 1/n$ or $\beta_n = 1/\sqrt{n}$ (theoretical guarantee)
- Adaptive: Choose β_n based on convergence progress

Visual Comparison: Iteration Schemes

Fixed Point Iteration Schemes

One-Step vs Two-Step Schemes

One-Step (Picard):

$$\boxed{x_{n+1} = Tx_n}$$

Two-Step (Ishikawa):

$$\boxed{y_n = (1-\alpha_n)x_n + \alpha_n Tx_n}$$

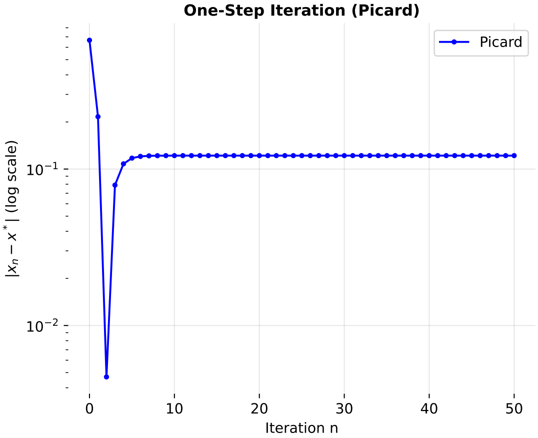
$$\boxed{x_{n+1} = (1-\beta_n)x_n + \beta_n Ty_n}$$

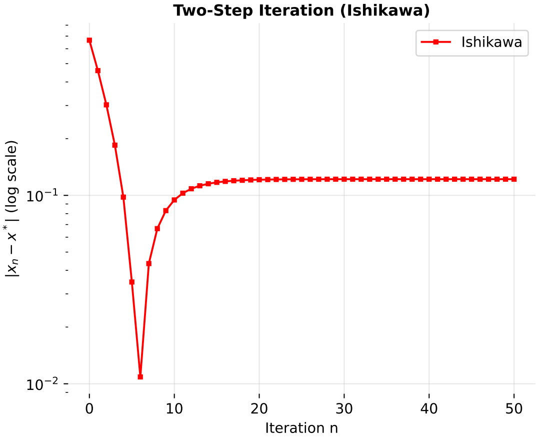
Convergence Characteristics

Picard:

Ishikawa:

Convergence Behavior: Numerical Example

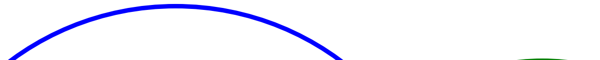




Nonexpansive Mapping Property

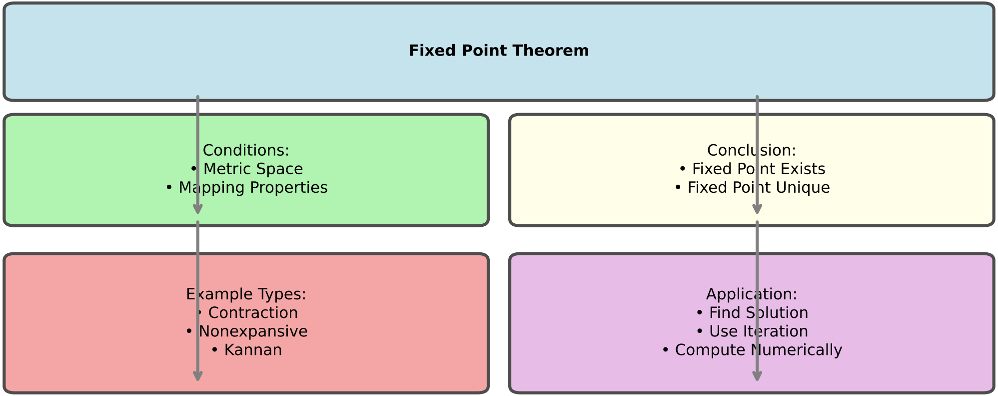
Nonexpansive Mapping Property

Nonexpansive Condition:
$\|Tx - Ty\| \leq \|x - y\|$ for all $x, y \in K$

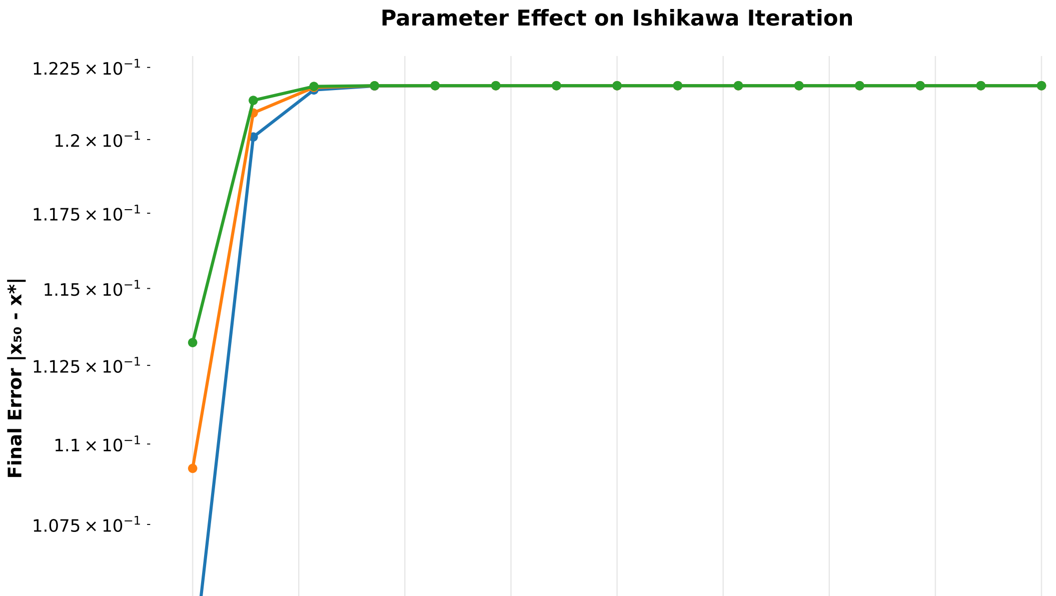


Fixed Point Theorem Structure

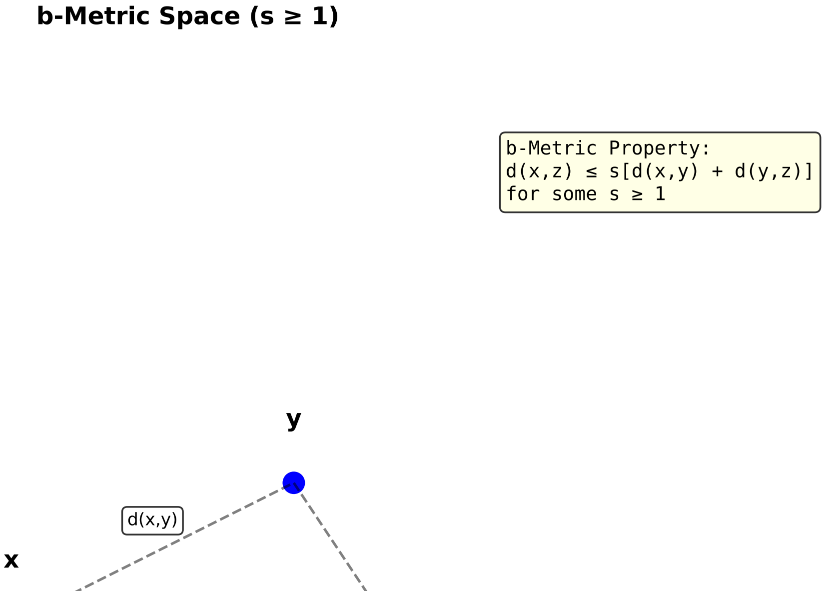
Fixed Point Theorem Framework



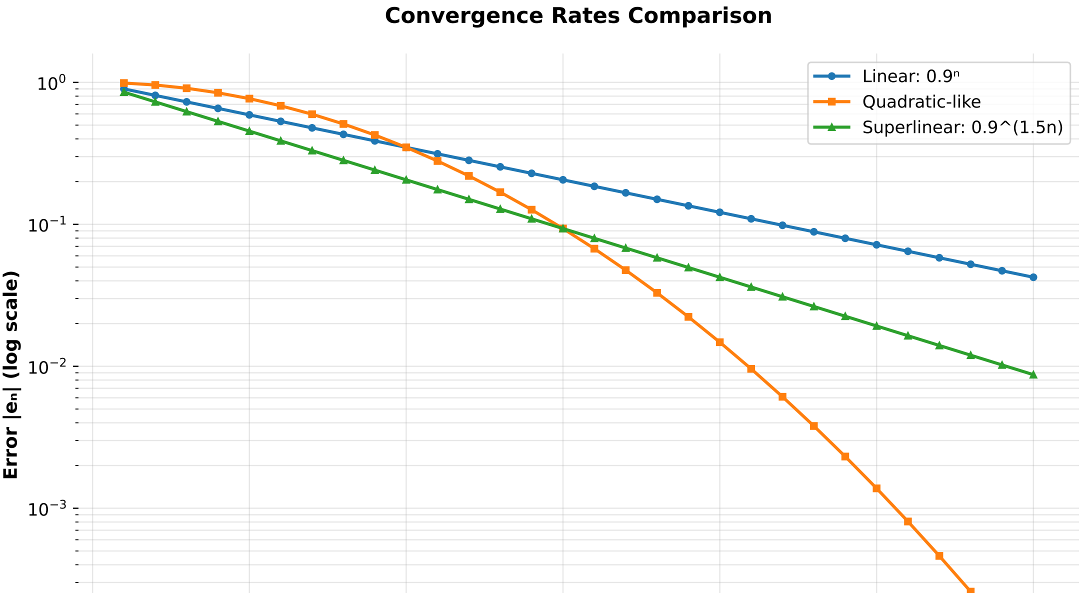
Parameter Sensitivity in Ishikawa Iteration



b-Metric Space Illustration



Convergence Rates Comparison



Python Implementation: Ishikawa Iteration

```
import numpy as np

def ishikawa(T, x0, alpha, beta, maxit=100):
  x = float(x0)
  for n in range(maxit):
    y = (1-alpha)*x + alpha*T(x)
    x_new = (1-beta)*x + beta*T(y)
    if abs(x_new - x) < 1e-8:
      return x_new, n
    x = x_new
  return x, maxit

x_star, iters = ishikawa(T, 0.1, 0.4, 0.6)
```

Common Fixed Point Problem (Theorem 5.132)

Solve system: $x - T_i(x) = f_i$ for $i = 1, \ldots, k$

Algorithm:

1. Define $F_i(x) = f_i + T_i(x)$ (nonexpansive)
2. Compute weighted operators: $U_r = (1-\alpha)I + \alpha F_r U_{r-1}$
3. Apply parameter-controlled iteration:

$$x_{n+1} = (1 - c_n)x_n + c_n F_k U_{k-1}(x_n)$$

   where $c_n \to 0$ and $\sum_n c_n(1 - c_n) = \infty$

Pseudocode:

```
for n = 1, 2, ...:
  Compute U_r = (1 − α)I + αF_r U_{r−1} for r = 1, ..., k
  Set c_n = 1/(n + 1)
  x_{n+1} ← (1 − c_n)x_n + c_n F_k U_{k−1}(x_n)
  if |x_{n+1} − x_n| < ϵ:  break
return x_{n+1}
```

Convergence: Weakly to a common fixed point of all F_i.

Key Applications

1. **Solving Operator Equations:**
    - $x - Tx = f$ (Lax-Milgram theory)
    - Nonlinear elliptic PDEs
    - Integral equations of Fredholm type
2. **Variational Inequalities:**
    - $\langle Ax - b, x - y \rangle \geq 0$ for all y in a convex set
    - Can be reformulated as fixed point problems
    - Applications in optimization and game theory
3. **Machine Learning:**
    - Federated learning with local and global updates (Ishikawa-type)
    - Consensus algorithms for distributed optimization
    - Deep learning: momentum-based optimization (related to two-step methods)
4. **Numerical Linear Algebra:**
    - Iterative methods for solving $Ax = b$
    - Eigenvalue problems
    - Matrix fixed points

Practical Considerations

When to use which method?

Picard iteration ($x_{n+1} = Tx_n$):

+ T is a contraction mapping ($k < 1$)
+ Need fast convergence
- Limited to certain mapping classes
- May diverge for nonexpansive mappings

Mann iteration:

+ T is nonexpansive
+ Want convergence guarantees
  Slower than Picard but more robust
+ Easier to implement than Ishikawa

Ishikawa iteration:

+ T is nonexpansive or quasi-nonexpansive
+ Need more control over iteration
+ Better stability properties
  Two steps per iteration (slower per step)
+ More parameters to tune

Main Results Summary

Fixed Point Theorems (Section 5.5):

- **Theorem 5.118:** Ćirić-Presić in b-metric spaces
- **Theorem 5.120:** Markov-Kakutani (affine mappings)
- **Theorem 5.123:** de Marr (nonexpansive families)
- **Theorems 5.126, 5.124-5.125:** Browder, Belluce-Kirk generalizations

Iteration Schemes (Section 5.5):

- **Theorem 5.130:** Strong convergence (compact strictly convex space)
- **Theorem 5.131:** Weak convergence (uniformly convex with Opial condition)
- **Theorem 5.132:** Parameter-controlled convergence
- Application to solving systems of operator equations

Contraction Sequences (Section 5.6):

- **Theorem 5.133:** Continuity of fixed points under contraction sequences
- **Theorem 5.127:** Multivalued Kannan-Reich type mappings

Key Conceptual Insights

1. **Generalization Hierarchy:**

$$\text{Contraction} \subset \text{Nonexpansive} \subset \text{Quasi-nonexpansive} \subset \text{General mappings}$$

2. **Iteration Method Progression:**

$$\text{Picard} \subset \text{Mann} \subset \text{Ishikawa} \subset \text{General Multi-step}$$

3. **Convergence Trade-offs:**
   - Stronger mapping classes $\Rightarrow$ stronger convergence (strong vs. weak)
   - More complex iteration $\Rightarrow$ wider applicability but slower per step
   - Parameter tuning can balance convergence rate and stability

4. **Space Geometry Matters:**
   - **Strict convexity:** Enables strong convergence
   - **Uniform convexity:** Stronger results, applies to more spaces
   - **Opial condition:** Allows weak convergence analysis

5. **Common Fixed Points vs. Individual Fixed Points:**
   - Finding common fixed point of family $\{F_i\}$ often easier than sequential finding

Research Directions and Open Problems

Extensions and Generalizations:

- Higher-order schemes (three-step, multi-step)
- Iteration for set-valued mappings
- Parallel and distributed iteration schemes
- Adaptive parameter selection strategies

Theoretical Questions:

- Rate of convergence (linear vs. superlinear)
- Optimal parameter choices
- Convergence in exotic spaces (non-reflexive, non-locally convex, etc.)

Computational Aspects:

- Efficient implementation on parallel/GPU hardware
- Error analysis and backward stability
- Preconditioning and acceleration techniques

Applications:

- Machine learning: convergence analysis of distributed methods
- Image processing: iteration schemes for inverse problems

Conclusion

Summary:

- Fixed point theorems provide existence and uniqueness results
- Iteration methods provide computational schemes to find fixed points
- Ishikawa iteration is a powerful two-step method applicable to nonexpansive mappings
- Parameter selection and space geometry play crucial roles

Takeaway Message:

"Two-step iteration methods like Ishikawa provide a powerful balance between generality and computational feasibility."

Reading Recommendations:

- Pathak: Chapters 5.5-5.7 on fixed point theorems
- Ishikawa (1974): Original paper on two-step iterations
- Browder (1965+): Classic work on nonexpansive mappings in Banach spaces
- Recent reviews on machine learning applications